Date
16 October 2026

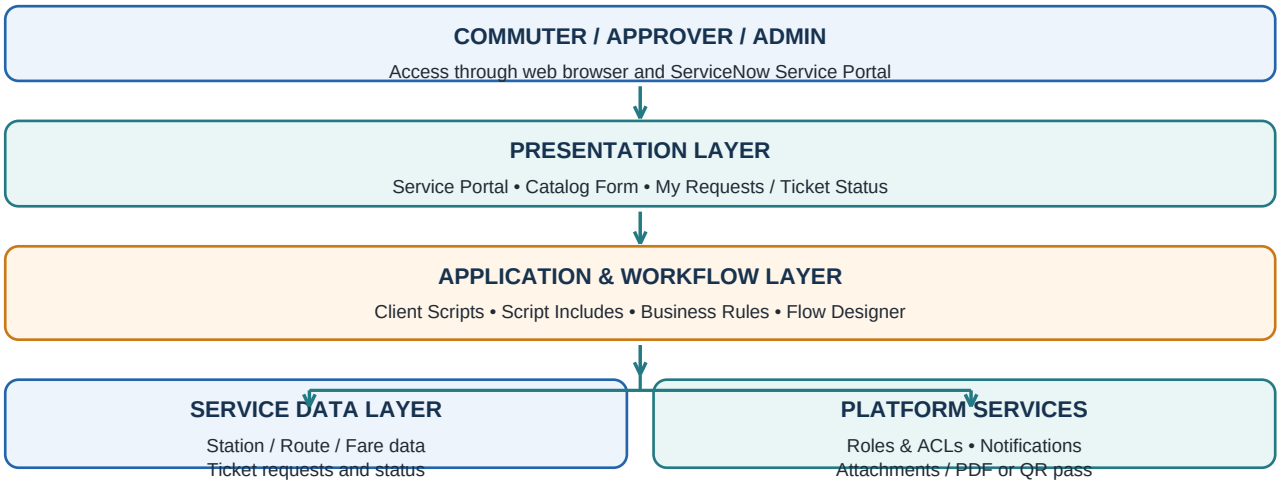
Project Name	Metro Ticket Generating System in ServiceNow	Maximum Marks	4 Marks

PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Technical Architecture

The proposed solution uses ServiceNow platform capabilities to support metro ticket booking, fare calculation, approval, ticket generation, delivery, and status tracking.



Hosted on the organization's ServiceNow cloud instance; no separate external API or ML model is specified in the current requirements.

Architecture flow: commuter or approver uses the Service Portal; platform scripts validate inputs and calculate fares; Flow Designer manages approval; approved requests generate a ticket and notification; records and status are maintained in ServiceNow.

Table 1: Components & Technologies

S.No	Component	Description	Technology
1	**User Interface**	Passenger-facing booking form, ticket summary, and request status view.	ServiceNow Service Portal and Service Catalog; browser-based UI.
2	**Application Logic-1**	Validate source and destination; capture travel date, ticket type, and passenger details.	Catalog Client Scripts and UI policies where appropriate.
3	**Application Logic-2**	Calculate fare, route distance, and ticket summary; recalculate when quantity or ticket type changes.	Script Includes and server-side JavaScript on ServiceNow.
4	**Application Logic-3**	Route requests to approvers, send notifications, and manage ticket state changes.	Flow Designer, Business Rules, and ServiceNow notifications.
5	**Database**	Store station lists, route mappings, fare rules, passenger request details, and ticket status.	ServiceNow platform tables / platform database.
6	**Cloud Database**	The project is hosted on ServiceNow; a separate cloud database is not specified in the requirements.	ServiceNow-managed platform data storage; no external database specified.
7	**File Storage**	Store or attach generated ticket summaries and PDF/QR-code ticket artefacts, if implemented.	ServiceNow attachments / platform file storage.
8	**External API-1**	No external API is identified in the supplied project requirements.	Not specified / not required by current scope.

S.No	Component	Description	Technology
9	**External API-2**	No second external API is identified in the supplied project requirements.	Not specified / not required by current scope.
10	**Machine Learning Model**	The project requirements do not specify a machine-learning feature.	Not applicable to the current scope.
11	**Infrastructure (Server / Cloud)**	Deploy and run the ticketing solution within the configured ServiceNow instance.	ServiceNow cloud platform; environment configuration depends on the project instance.

PROJECT DESIGN PHASE-II

Technology Stack (Architecture & Stack)

Table 2: Application Characteristics

S.No	Characteristics	Description	Technology
1	**Open-Source Frameworks**	The supplied requirements describe a ServiceNow-native implementation and do not name an open-source framework.	ServiceNow platform features; no separate open-source framework specified.
2	**Security Implementations**	Use ServiceNow authentication and role-based access control for Commuters, Approvers, and Admins. Protect fare tables and passenger identity data from unauthorized access.	ServiceNow roles, Access Control Lists (ACLs), authentication, and platform security controls.
3	**Scalable Architecture**	Logical separation of presentation, application/workflow logic, and data supports maintainable platform configuration. Actual capacity depends on the ServiceNow instance and configuration.	ServiceNow platform architecture; Service Portal, scripts, Flow Designer, and platform tables.
4	**Availability**	The ticket booking portal is intended to be available for commuter use. Availability targets and service guarantees must follow the organization's ServiceNow service agreement.	ServiceNow cloud hosting and platform availability features; no separate load balancer is specified.
5	**Performance**	The project requirement states that fare summaries should update within 1.5 seconds of user input on the Service Portal form. Test scripts and form behaviour against this target.	Client Scripts for responsive validation; Script Includes for fare logic; efficient table queries and server-side processing.

Project Scope Notes

This technology stack is based on the uploaded Technology Stack template and the provided Metro Ticket Generating System in ServiceNow requirements. The source requirements specify booking, dynamic fare calculation, approvals, ticket generation and delivery, status tracking, cancellation, security, availability, scalability, and performance. External APIs and machine-learning components are not specified, so they are marked as not required or not specified rather than assumed.

Project Name: Metro Ticket Generating System in ServiceNow